

The Audenaert–Datta and Carlen–Frank–Lieb Conjectures: Literature Answer by Zhang

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Source questions

Carlen, Frank, and Lieb, arXiv:1806.03985, formulate on PDF page 7 (Conjecture 1) the Audenaert–Datta conjecture: for $1 \leq p \leq 2$, $-1 \leq q < 0$, and any $K \in \mathcal{M}_N$,

$$(A, B) \mapsto \mathrm{Tr} \left(B^{q/2} K^* A^p K B^{q/2} \right)^{1/(p+q)}$$

should be jointly convex on $\mathcal{P}_N \times \mathcal{P}_N$. The singular corner $(p, q) = (1, -1)$ is implicitly excluded because $1/(p+q)$ is not defined.

On PDF page 10, Conjecture 4 asks for the stronger statement that

$$\Psi_{p,q,s}(A, B) = \mathrm{Tr} \left(B^{q/2} K^* A^p K B^{q/2} \right)^s$$

is jointly convex throughout the same parameter range whenever $s \geq 1/(p+q)$.

Answering paper

Haonan Zhang, arXiv:1811.01205, explicitly reproduces the Audenaert–Datta conjecture as its Conjecture 1 and the stronger Carlen–Frank–Lieb conjecture as its Conjecture 2 on PDF page 2. The paper then says that it confirms the stronger conjecture.

More precisely, Theorem 1.1(3) on PDF page 3 proves that, for every invertible K ,

$$-1 \leq q \leq 0, \quad 1 \leq p \leq 2, \quad (p, q) \neq (1, -1), \quad s \geq \frac{1}{p+q}$$

implies joint convexity of $\Psi_{p,q,s}$. A remark immediately after the introduction explains that arbitrary, possibly singular K follow by approximating K with invertible matrices and using stability of convexity under pointwise limits.

Thus Zhang proves Conjecture 4 of arXiv:1806.03985 in full. Taking $s = 1/(p+q)$ proves Conjecture 1, the Audenaert–Datta conjecture. The same paper also states that this supplies the full parameter range for the data-processing inequality of the α - z Rényi divergences.

Status

This is an explicit later-literature affirmative answer, not a new proof from the present run. Both source conjecture signals are closed by the stronger Theorem 1.1(3) of arXiv:1811.01205.

References

- E. A. Carlen, R. L. Frank, and E. H. Lieb, *Inequalities for quantum divergences and the Audenaert–Datta conjecture*, arXiv:1806.03985.
- H. Zhang, *From Wigner–Yanase–Dyson conjecture to Carlen–Frank–Lieb conjecture*, arXiv:1811.01205, Theorem 1.1(3).